Lecture - 7

❖ Testing and Debugging:

Testing and Debugging are important processes in software engineering. Testing and Debugging seems to be similar but they are quite different. The main difference between Testing and Debugging are testing is the process of identifying defects or errors in the software system while debugging is the process to fix those defects or errors that affect the working of software.

➢ Testing:

Testing is the process of checking a program to determine whether it produces the expected output for different inputs.

- ❒ ***Purpose of testing:***
  - ✓ Find bugs in the program.
  - ✓ Check whether the program meets requirements.
  - ✓ Verify that the output is correct.
  - ✓ Check the program with normal, boundary, and invalid inputs.
- ❒ ***Example:***

  Suppose a program calculates the average of two numbers.

  Input:

  10, 20

  Expected output:

  15

  If the program produces 15, the test passes. If it produces 20, there is likely a problem that needs debugging.

➢ Debugging:

Debugging is the process of finding, analyzing, and fixing bugs in a program.

- ❒ ***Typical debugging steps:***

  Program → Test → Find Bug → Identify Cause → Fix Bug → Test Again

- ❒ ***Example:***

```
int a = 10, b = 20;
printf("%d", a - b);
```

If the programmer intended to add the numbers, the output is -10 instead of 30.
Bug: Wrong operator (- instead of +)
Fix:

```
printf("%d", a + b);
```

- **Difference between Testing and Debugging**

Testing	Debugging
Finds whether a program has problems	Finds and fixes the cause of the problem
Detects bugs	Corrects bugs
Usually performed by testers/programmers	Mainly performed by programmers